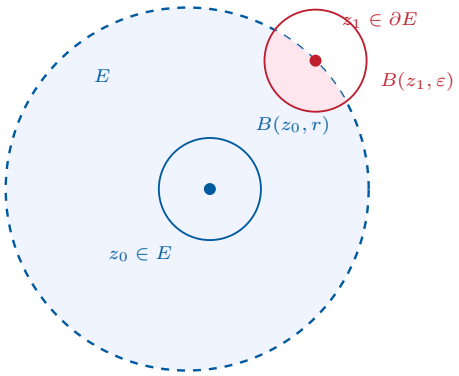
复变函数答案

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1　习题一

16. 证明:



粉色区域为 $B(z_1,\varepsilon) \cap E \neq \varnothing$;
而 $B(z_0,r) \subseteq E$

一、$\partial E \cup E \subseteq E'$

1° $E \subseteq E'$:
设 $z_0 \in E$.
$\because E$ 为开集,
$\therefore \exists r > 0$ 使 $B(z_0, r) \subseteq E$.
$\forall \varepsilon > 0$, 取 $w = z_0 + \frac{1}{2}\min(r, \varepsilon)$,
则 $0 < |w - z_0| < \min(r, \varepsilon)$,
$\therefore w \in E$ 且 $w \in B(z_0, \varepsilon) \setminus \{z_0\}$,
$\therefore z_0 \in E'$, 即 $E \subseteq E'$.

2° $\partial E \subseteq E'$:
设 $z_0 \in \partial E$.
若 $z_0 \in E$, 由 1° 知 $z_0 \in E'$;
若 $z_0 \notin E$,
$\because z_0 \in \partial E$,
$\therefore \forall \varepsilon > 0, B(z_0, \varepsilon) \cap E \neq \varnothing$,
且其中各点均异于 z_0,
$\therefore \big(B(z_0, \varepsilon) \setminus \{z_0\}\big) \cap E \neq \varnothing$,
$\therefore z_0 \in E'$.
综上, $\partial E \cup E \subseteq E'$.

二、$E' \subseteq \partial E \cup E$

设 $z_0 \in E'$.
若 $z_0 \in E$, 结论已成立;
若 $z_0 \notin E$,
$\because z_0 \in E'$,
$\therefore \forall \varepsilon > 0, \big(B(z_0, \varepsilon) \setminus \{z_0\}\big) \cap E \neq \varnothing$,
$\therefore B(z_0, \varepsilon) \cap E \neq \varnothing$;
又 $\because z_0 \notin E$,
$\therefore z_0 \in B(z_0, \varepsilon) \cap E^c$,
$\therefore B(z_0, \varepsilon) \cap E^c \neq \varnothing$,
$\therefore z_0 \in \partial E$.
综上, $E' \subseteq \partial E \cup E$.

综合一、二, $E' = \partial E \cup E$. ■